

Hand-in assignment 3

Instructions

- This assignment consists of **two questions**.
- On Vula it opens at 12:00 on Friday, 19 April 2024, please upload your solutions under **Vula Assignments no later than 23:55 on Tuesday, 23 April 2024**. Make sure that no pages or questions are missing.
- Write your full name and student number on your solutions.
- If possible, **upload a single document** saved as **ABCXYZ001_A3** (where **ABCXYZ001** is your campus Id).
- Show all your work or reasoning. **Answers without work or reasoning shown where needed will receive no mark.**

Questions

1. State whether the following are true or false. If true, prove it or if false, give a counter-example.
 - a) Let (X, d) be a metric space and $(A_i)_{i \in I}$ a family of complete subsets of X , then $\bigcup_{i \in I} A_i$ is complete. [3]
 - b) Let (X, d) be a metric space and $(A_i)_{i \in I}$ a family of complete subsets of X , then $\bigcap_{i \in I} A_i$ is complete. [3]
 - c) Let (X, d_X) , (Y, d_Y) be metric spaces, A a complete subset of X and $f : X \rightarrow Y$ a continuous function, then $f(A)$ is a complete subset of Y . [3]
 - d) Let (X, d_X) , (Y, d_Y) be complete metric spaces, B a complete subset of Y and $f : X \rightarrow Y$ a continuous function, then $f^{-1}(B)$ is a complete subset of X . [3]
 - e) Let (X, d) be a complete metric space and $(A_n)_{n \in \mathbb{N}}$ a strictly decreasing sequence of subsets of X , then $\bigcap_{n \in \mathbb{N}} \overline{A_n} \neq \emptyset$. [3]
2.
 - a) Let (Y, d_Y) be a metric space and (Z, d_Z) a complete metric space, suppose (X, d_X) is a dense subspace of (Y, d_Y) and $f : X \rightarrow Z$ a uniformly continuous function. Prove that there exists a uniformly continuous function $g : Y \rightarrow Z$ such that $g(x) = f(x)$ for all $x \in X$. [5]
 - b) Give an example to show that completeness of (Z, d_Z) is needed in the previous question. [3]